

Mini Fridge With PELTIER MODULES

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It was in the middle of 1821 when J.T. Seebeck discovered that if two dissimilar metals connected at two different points are held at different temperatures, a microvoltage develops. This phenomenon is called Seebeck effect. Some years later, Peltier discovered that if a voltage is applied to a thermocouple, one junction of the thermocouple heats up while the other cools down. Opposite of the Seebeck effect is called the Peltier effect.

This guide to designing a little solid-state cooler engine is centered on the commonly available Peltier chip. A Peltier chip is a thermo-element that utilises the Peltier effect to implement a heat pump. It has two plates, one cold and the other hot. Between the plates there are several thermocouples connected together. If a proper voltage is applied, one plate becomes cold and the other plate becomes hot.

The Peltier chip is called a heat pump because it neither generates heat nor cold. It just transfers heat from one plate to another, thus cooling the first plate. It is also popularly called thermoelectric cooler (TEC) chip. In short, by applying a direct current (DC) to the TEC chip, temperature differences are generated between the front and back of the device (Peltier effect), and you end up with a hot and a cold surface. TEC1-12706 is a common thermoelectric cooler chip available from most eBay traders.

In TEC1-12706, the letter C after TE indicates 'standard size,' and 1 indicates 'one-stage' TEC. Right next comes a dash. After the dash, the first three digits indicate the number of thermocouples inside the TEC. Here, it has 127 couples. Next two numbers



Fig. 1: Key components for DIY Peltier cooler

In short, when DC voltage is applied to the TEC chip, positive and negative charge carriers in the pellet array absorb heat energy from one substrate surface and release it to the substrate at the opposite side. The surface where heat energy is absorbed becomes cold, and the opposite surface where heat energy is released becomes hot!

The Peltier cooler also includes a powerful heat-sink/fan combination to cool the TEC chip. Table below depicts specifications of a TEC1-12706 thermoelectric cooler chip. You can buy a CPU heat-sink and fan unit with almost the same specifications as the CPU fan for AMD processors: 80.6 × 80.6 × 69.4mm³ with aluminium fin heat-sink. The additional 60 × 60mm² aluminium heat-sink plate (and thermal grease) is also available at reasonable price. Fortunately, you can buy most of these key components from prominent eBay and/or Amazon sellers (see Fig. 1).

SPECIFICATIONS OF 40×40mm ² TEC1-12706		
	Min.	Max.
Hot-side temperature (°C)	25	50
Q _{max} (watts)	50	57
Delta T _{max} (°C)	66	75
I _{max} (amps)	6.4	6.4
V _{max} (volts)	14.4	16.4
Module resistance (ohms)	1.98	2.30

indicate operating current rating for the Peltier. So, 06 denotes '6 amperes.'

Peltier cooler

A Peltier cooler is a cooler engine comprising a Peltier element (TEC chip). When a direct current is passed through a TEC chip, the low-temperature side absorbs heat and the high-temperature side emits heat, creating a temperature difference across the two surfaces. However, since the heat emitted is more reactive to the amount of electricity input into the module than the heat absorbed, if a direct current is continuously passed through the chip the emitted heat exceeds the absorbed heat and both sides of the unit become hot. Because of this, it is crucial to connect the TEC chip to a radiator such as aluminium fins to efficiently disperse the emitted heat.

TEC chip and basic test

Before starting an actual construction with a TEC chip, test it for proper working condition. For that, just connect the red (+) and black (-) wires of the TEC chip (TEC1-12706) to a 1.5V DC lab power supply and keep the power supply on for 10 to 30 seconds. Thereafter you can test the TEC chip using your fingertip or digital thermometer to ensure that one side of the chip is hot and the other side cold. Just mark down hot and cold faces of the TEC chip (for example, with letters H and C) using any permanent marker pen.

Powering up

The assembled cooler engine (thermoelectric cooler chip, heat-sink and